Bootstrap Grid System (Bootstrap 5)

The **Grid System** in Bootstrap is a **flexbox-based responsive layout system** used to create structured, mobile-first web page layouts.

It is built on a **12-column system** and uses predefined CSS classes.

1.Why Bootstrap Grid?

Before Bootstrap, developers wrote custom CSS with floats and media queries.

Bootstrap Grid provides:

- ✅ Responsive design (mobile → tablet → desktop)
- ✅ Predefined breakpoints
- ✅ Flexbox-based alignment
- ✅ Fast layout development
- ✅ Consistent structure

2.Core Grid Structure

Bootstrap grid always follows this structure:

```
<div class="container">
  <div class="row">
    <div class="col">Column</div>
  </div>
</div>
```

- **3 Main Components**

Class	Purpose
container	Main wrapper
row	Horizontal group of columns
col	Actual column

3.Container Types

1. .container

Responsive fixed width container.

2. .container-fluid

Full width (100% width).

3. .container-{breakpoint}

Responsive container that becomes full-width until specific breakpoint.

Example:

```
<div class="container-md">
  Content
</div>
```

4.12-Column System

Bootstrap divides each row into **12 equal columns**.

Example:

```
<div class="container">
  <div class="row">
    <div class="col-4">4 Columns</div>
    <div class="col-4">4 Columns</div>
```

```
  <div class="col-4">4 Columns</div>
 </div>
</div>
```

👉 4 + 4 + 4 = 12

5.Breakpoints (Responsive Design)

Bootstrap Grid uses 6 breakpoints:

Breakpoint	Class Prefix	Screen Size
Extra small	col-	<576px
Small	col-sm-	≥576px
Medium	col-md-	≥768px
Large	col-lg-	≥992px
Extra Large	col-xl-	≥1200px
XXL	col-xxl-	≥1400px

Example: Responsive Layout

```
<div class="row">
  <div class="col-12 col-md-6 col-lg-4">Column</div>
  <div class="col-12 col-md-6 col-lg-4">Column</div>
  <div class="col-12 col-md-6 col-lg-4">Column</div>
</div>
```

Behavior:

- 📱 Mobile → Full width (12)
- 💻 Tablet → 6 columns each

- 🖥️ Desktop → 4 columns each

6.Auto Layout Columns

If you just write:

```
<div class="row">
  <div class="col">1</div>
  <div class="col">2</div>
  <div class="col">3</div>
</div>
```

Bootstrap automatically divides into equal columns.

---

7. **Column Wrapping**

If total columns > 12 → next column moves to next line.

Example:

```
<div class="row">
  <div class="col-8">8</div>
  <div class="col-6">6</div>
</div>
```

👉 8 + 6 = 14 → second column wraps to next line.

8.Offsetting Columns

Used to create space before a column.

```
<div class="row">
  <div class="col-md-4 offset-md-4">Centered Column</div>
</div>
```

9.Nesting Columns

You can create a row inside a column.

```
<div class="row">
  <div class="col-8">
    <div class="row">
      <div class="col-6">Nested 1</div>
      <div class="col-6">Nested 2</div>
    </div>
  </div>
</div>
```

---

10. Alignment Utilities (Flexbox)

Bootstrap grid uses Flexbox internally.

Horizontal Alignment

```
<div class="row justify-content-center">
```

Options:

- start
- center
- end
- between
- around
- evenly

Vertical Alignment

```
<div class="row align-items-center">
```

Options:

- start

- center
- end

---

11. Gutters (Spacing Between Columns)

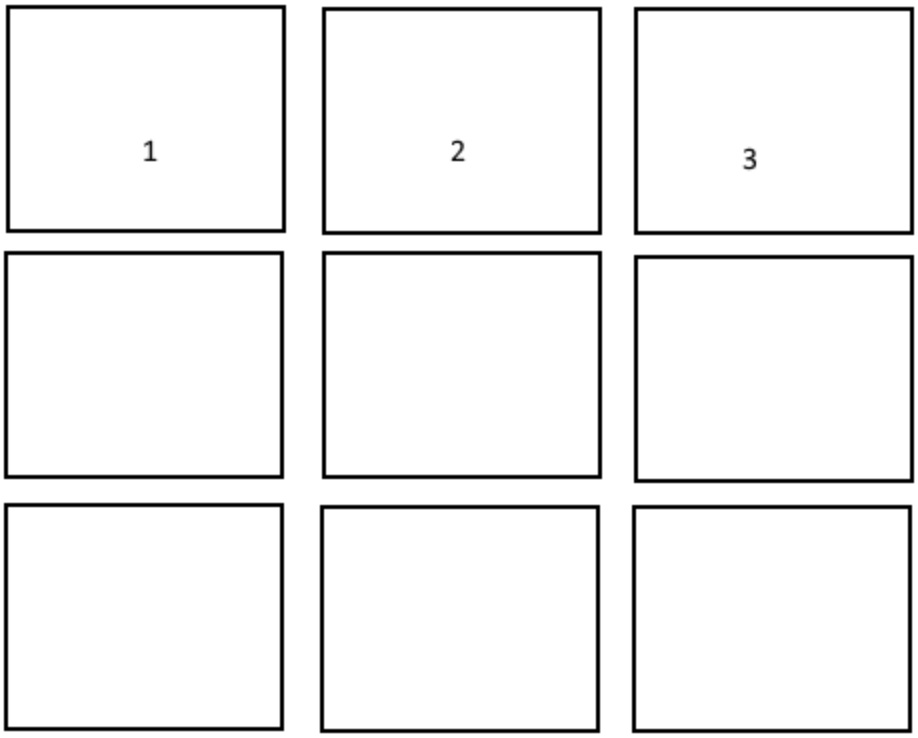
Default gutter = padding inside columns.

Change gutter:

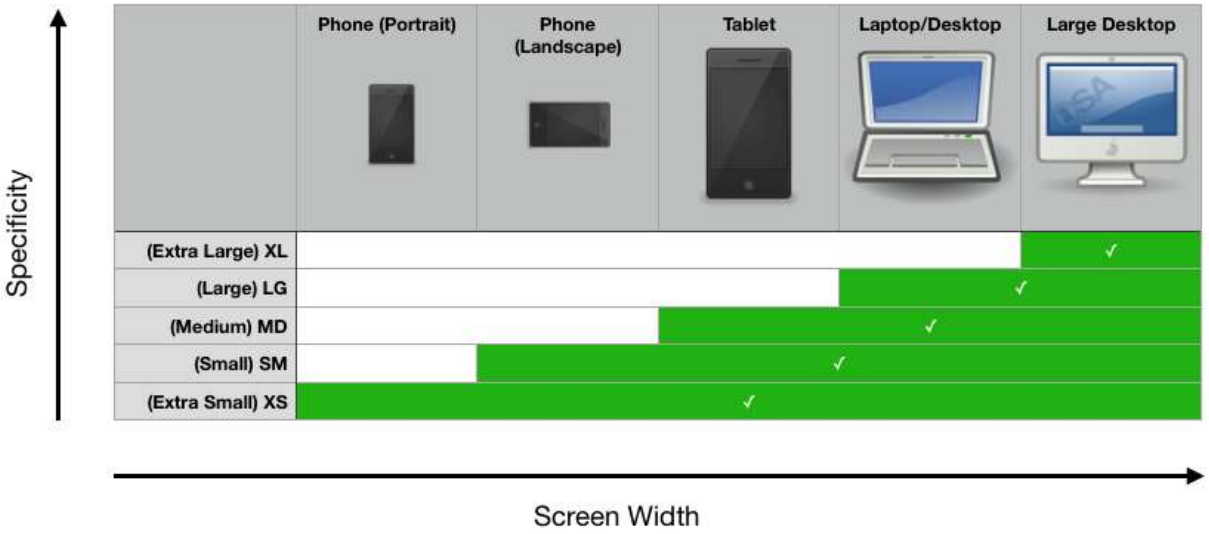
Options:

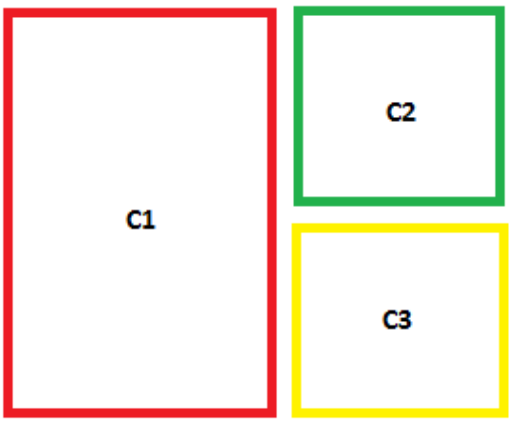
- g-0 → no spacing
- g-1 to g-5

12.Visual Understanding of Grid

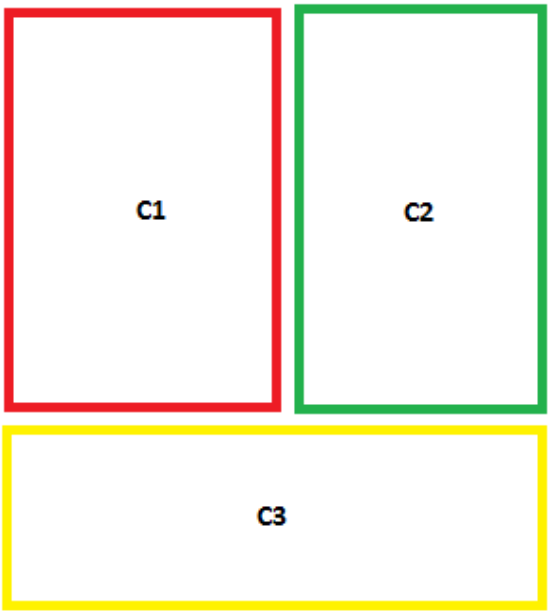


Bootstrap Breakpoints





max-width: 1200px (desktop)



max-width: 768px (mobile)